

Review Protocol

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Review Rationale

- There is an increasing number of ML algorithms on embedded systems. Driven by TinyML, which allows the execution of ML models on resource-constrained devices like Microcontroller Units (MCUs) or small Single Board Computers, this integration aims to reduce latency, strengthen data privacy, lower power consumption, and reduce connectivity dependencies
- Many application domains are required to fulfill the quality attributes that TinyML enables; various algorithms are distributed across different microcontrollers in industrial plants or sites. Thus, Lifecycle Management of such models, encompassing deployment, monitoring, updates, and eventual retirement, is a topic of high relevance, only gaining in interest with increasing deployment of sensors and TinyML models for data acquisition.
- Existing industry solutions like Edge Impulse are used for cloud development and edge deployment, without addressing any other aspects that are part of MLOps, e.g., monitoring, updates, and eventual retirement
- Other solutions like AWS Greengrass allow decoupling of edge devices for a certain time, but still are cloud-driven and not designed for on-device adaptation
- Thus, to address individual requirements like an autonomous TinyMLOps system, an overview is necessary to illustrate the architecture, frameworks, and tools of existing solutions to address individual requirements for implementation of TinyMLOps
- Existing reviews do not cover the architecture, frameworks, and tools of existing solutions:
 - Rani et al. (2024): Explore Edge MLOps from an industrial perspective; Cover: key aspects, procedures and challenges; Highlights: Emphasize that resource constraints, connectivity, and model evolution are key challenges → *None of the relevant aspects covered*
 - Rocha et al. (2024): Explore ML operation on medical devices; Cover: Distributed architectures, techniques and strategies; Limitations: Only focus on the medical domain, lacks comparison of existing solutions to ease understanding and enable integration into own solution stack → *Relevant aspects partially covered, limited and only investigated in medical domain*
 - Somvanshi et al. (2025): Explore the development of Deep Learning Solutions for TinyML; Cover: Development methods, used hardware, frameworks for on-device learning; Limitations: MLOps Lifecycle aspects not covered, only on-device

learning, architecture is not a real focus, only development methods → *Relevant aspects partially covered, limited and lacking the “bigger” architectural picture*

- Murshed et al. (2021): Explore development of TinyML and tools, solution stacks, covered use cases, and encountered challenges to make ML operation on resource-constrained devices possible; Limitations: MLOps Lifecycle aspects not covered, architecture is not a real focus, only development methods → *Relevant aspects partially covered, limited and lacking MLOps and architectural connection*
- Chadli et al. (2024): Explore methodologies, tools, and metrics to enhance sustainability of ML-based systems in MLOps; Limitations: provide strategies for environmental impact mitigation, architecture is not a real focus, frameworks are not covered, only mapping → *Relevant aspects partially covered, limited and lacking architectural connection*
- Morariu et al. (2024): Explore MLOps within the maritime domain to map algorithms, tools, and practices for MLOps; Limitations: Only partially addresses architectural aspects; focus is on the maritime domain only → *Relevant aspects partially covered, limited and only investigated in medical domain*

Research Questions

Overall goal: identify the state-of-the-art of MLOps implemented in resource-constrained systems.

RQ1

Aim: Create a big-picture of existing architectural approaches

What system architectures, methodologies, and practices are employed to support MLOps implementation in resource-constrained embedded systems?

Kitchenham Question Type: assessing frequency/rate of architectural-practice adoption

RQ2

Aim: Based on the architectural approaches, identify existing frameworks and tools to reduce cloud reliance for autonomous TinyMLOps

Which frameworks, tools and strategies facilitate decentralized on-device MLOps while minimizing reliance on cloud infrastructure?

Kitchenham Question Type: identifying technology/strategy options

Search Strategy

Overall Search Strategy:

- Snowballing, as per Wohlin's guideline (Update from Wohlin highlights that hybrid search should be performed → Include DB-Backup for seed identification)
- Use the guideline of Kitchenham complementary

Steps:

1. Seed identification based on Google Scholar
 - Identification of relevant search terms that describe our field
 - Define search string
 - Identification of the most relevant work (up to 10)
 - Based on studies found, create inclusion criteria that are met only by the most relevant studies
2. Seed identification based on databases (to back up the manual search)
 - Databases:
Selected the following to ensure the inclusion of high-quality research papers and cover the most relevant technical/engineering databases that likely hold papers of interest
 - i. ACM Digital Library: 3
 - ii. SpringerLink: 1
 - iii. IEEE Xplore: 4
 - iv. ScienceDirect: -
 - Use the defined search string to identify the most relevant work that meets ic
 - If necessary: Adapt string based on terms in studies
3. Receive final seed set
4. Forward and Backward Snowballing
 - Stopped after 2 iterations: No new relevant studies
 - Initial Search: Lukas, Backup search: Philipp, QS: Marco
5. Forward Snowballing based on initial result set for update
 - Due to time-consuming analysis, paper writing and fast-growing field
 - Update search: Philipp, QS: Marco

No grey literature inclusion: Although grey literature may include additional findings or “newer” developments in the field, its quality and trustworthiness cannot be assessed and is therefore not included in the search. This ensures the study's results are trustworthy so others can build their own solutions based on them.

General String:

(TinyML OR "Tiny Machine Learning" OR "Embedded Machine Learning" OR "On-device Machine Learning")

AND

(MLOps OR "Machine Learning Operations" OR "Lifecycle Management" OR Operationalization OR "Continuous Training" OR "Model Monitoring" OR TinyMLOps)

Strings per DB:

DB	String
Scholar	TinyML "Tiny Machine Learning" "Embedded Machine Learning" "On-device Machine Learning" MLOps "Machine Learning Operations" "Lifecycle Management" "Operationalization" "Continuous Training" "Model Monitoring" TinyMLOps
ACM	(TinyML OR "Tiny Machine Learning" OR "Embedded Machine Learning" OR "On-device Machine Learning") AND (MLOps OR "Machine Learning Operations" OR "Lifecycle

	Management" OR Operationalization OR "Continuous Training" OR "Model Monitoring" OR TinyMLOps)
Springer	(TinyML OR "Tiny Machine Learning" OR "Embedded Machine Learning" OR "On-device Machine Learning") AND (MLOps OR "Machine Learning Operations" OR "Lifecycle Management" OR Operationalization OR "Continuous Training" OR "Model Monitoring" OR TinyMLOps)
IEEE	("Full Text & Metadata":TinyML OR "Full Text & Metadata": "Tiny Machine Learning" OR "Full Text & Metadata": "Embedded Machine Learning" OR "Full Text & Metadata": "On-device Machine Learning") AND ("Full Text & Metadata": MLOps OR "Full Text & Metadata": "Machine Learning Operations" OR "Full Text & Metadata": "Lifecycle Management" OR "Full Text & Metadata": Operationalization OR "Full Text & Metadata": "Continuous Training" OR "Full Text & Metadata": "Model Monitoring" OR "Full Text & Metadata": TinyMLOps)
Science Direct (search string had to be separated due to limit)	(TinyML OR "Tiny Machine Learning" OR "Embedded Machine Learning" OR "On-device Machine Learning") AND (MLOps OR "Machine Learning Operations" OR "Lifecycle Management" OR Operationalization) (TinyML OR "Tiny Machine Learning" OR "Embedded Machine Learning" OR "On-device Machine Learning") AND ("Continuous Training" OR "Model Monitoring" OR TinyMLOps)

Selection Procedures

IC:

1. The study addresses TinyML concepts, methodologies, or challenges
2. The study presents practical implementations, experiments, or real-world case studies in the TinyML domain, relevant for the operation of models
3. The study addresses lifecycle management characteristics of AI-based applications on embedded devices

EC:

1. The study does not meet basic scientific standards (e.g., clear methodology, reproducibility, logical reasoning in conclusions)
2. The study is not peer-reviewed or appears in non-reputable sources
3. The study does not contribute original research or data (e.g., meta-analyses, secondary reviews, purely conceptual discussions without validation)
4. The study is published before 2020
5. The study is not in English

Baseline Selection and Criteria Application:

In his guideline (originally from EASE'14), Wohlin recommends applying the criteria for snowballing when reviewing the reference list of each paper. This results in a baseline exclusion being carried out based on language, publication year, and type of publication. In our case, we downloaded the paper and crossed out the relevant references in the PDF. Building on this, the criteria are checked against the title and the author. For example, a paper may be included even if the title does not suggest it should be: for instance, if the author is already known to likely make valuable contributions to the field of interest. For the remaining papers, a screening of the abstracts and relevant passages of the paper is then conducted to make a final decision, while checking the references. Therefore, information regarding exclusions is not explicitly documented in a spreadsheet—as in a traditional database-based review where a spreadsheet export can be generated—but is noted directly within the paper itself by crossing it out. The spreadsheet only documents which papers were excluded following this reference screening: This can occur, for example, when results are being extracted, and it becomes apparent that a specific inclusion criterion was not met after all. The inclusion and exclusion criteria were continuously reviewed throughout the entire review process so that adjustments could be made if necessary. This, however, did not occur.

QS Procedure:

The first researcher (Lukas) applies the procedure for “Baseline Selection and Criteria Application” after each iteration, a spreadsheet is shared with the backup (Philipp). The backup conducts the same procedure for “Baseline Selection and Criteria Application” with the ability to override within the same spreadsheet. Finally, the resulting spreadsheet is handed to the third researcher (Marco) for QS. The spreadsheet is filled iteratively to produce the initial result set.

Update Procedure:

Since the field is fast-developing, new papers are published on an ongoing basis. To cope with this, after data extraction and paper writing, we took the initial result set and conducted one extra iteration of forward snowballing. For complementation, another manual search using Google Scholar based on the defined search string is executed. The same procedure for “Baseline Selection and Criteria Application” was applied by Philipp. Following this, the resulting spreadsheet is handed to Marco for QS. The resulting spreadsheet is the final result set.

Inter-rater agreement: Not documented, since the backup had the purpose of quality assurance and refinement. The final QS (Marco) checked these decisions.

Quality Assessment of Studies

Quality assessment is applied in four steps:

1. Baseline check during selection for IC/EC (done by two researchers), here:
 - non-peer-reviewed papers are sorted out

- non-reputable sources, e.g., papers that can only be retrieved via .ru or http pages
- papers that do not meet basic scientific standards, e.g., without clear methodology or with gaps in the research design, are sorted out
- 2. Final baseline check during selection for IC/EC by a third researcher for QS
- 3. Application of the RTF/CTF classification (as a substitute for Kitchenham's QS checklist, RTF/CTF combination used as indicator for quality and maturity)
 - Detailed inspection of the studies according to definitions
 - Studies where this is not possible, or the lack of quality becomes evident, would be sorted out (did not happen)
- 4. Check of the RTF/CTF classification by the third researcher for QS

Comment: The Quality Checklist by Kitchenham (Tables 5 & 6) was applied only within the context of EC1 and was not explicitly documented due to the procedure (see Selection Procedure). Many aspects of the Quality Checklist get lost in details that are irrelevant to our analysis. Furthermore, the field is not yet mature enough for this checklist to be fully applied. Therefore, we applied RTF/CTF classification as a substitute for Kitchenham's QS checklist. In general, we consider the combination of EC1, EC2, RTF/CTF classification, and quality assurance by a third researcher sufficient to ensure a high quality of the included studies. This is further ensured by exclusively considering peer-reviewed studies as a fundamental baseline for quality assurance.

Data Extraction

Extraction was applied in two steps:

1. Context and Metadata Analysis
 - Based on title and abstract; complemented by full text if details are missing
 - Extraction of:
 - i. Objective
 - ii. Scope
 - iii. Key results
 - iv. Identified gaps
 - v. Application domain (Use Case)
 - vi. AI-related task
 - vii. RTF
 - viii. CTF
2. Data Extraction and Analysis
 - Based on full text
 - Category system for extraction of architectural details initially created based on the seed set, then inductively evolved across the full initial result set, re-evaluated (unchanged) against the full final result set from the update
 - Extraction of:
 - i. General Architecture
 1. Styles
 2. Frameworks
 3. Tools
 4. Applied technologies
 5. Runtime details

- 6. Monitoring metrics
 - 7. Covered MLOps Phases
- ii. Patterns
 - 1. Training
 - 2. Management
 - 3. Learning
- iii. Tactics
 - 1. Model Optimization
 - 2. CI/CD

All extracted information was stored in a spreadsheet per study: identification number, search step, paper metadata (author/ref/year/publication type/database), inclusion/exclusion decision, context metadata, general architecture details, patterns, tactics

Extraction Procedure: Similar to the selection procedure; therefore, no interrater agreement. Extraction was carried out by the first researcher, then the spreadsheet was handed to the second researcher. The second researcher conducted his own extraction similar to the first researcher, with the ability to override within the spreadsheet. Finally, the resulting spreadsheet was handed to the third researcher (Marco) for QS. The spreadsheet is filled iteratively to produce the initial result set, i.e., for each iteration of snowballing, phases are applied like: Selection → Quality Assessment → Extraction (should prevent later exclusion and tracing of studies based on references that have been wrongly passed to extraction, which would result in large overhead and effort)

Detection of duplicates via DOI or, if not available, title in the spreadsheet via an automatic formula.

While the data extraction is conducted by three researchers, as described above, the overall result of the extracted data is going to be quality assured by a fourth researcher (Christian), called in prior to the synthesis of the results.

Synthesis Approach

RQ1

Objective for RQ1: Identify research trends, patterns and knowledge gaps → Rather descriptive/structural

Methodical Approach: classification and mapping techniques borrowed from Petersen's mapping-study instrument → Creation of multi-dimensional maps to enable analysis of results not apparent from a single or two dimensions.

RQ2

Objective for RQ2: Cluster frameworks, tools and extract their strategies to reduce cloud reliance

Methodical Approach: Extraction and clustering of Framework purposes, including features and strategies for reducing cloud reliance; clustering of tools according to MLOps phases to distinguish between tools, highlight competing or complementary tools, and enable practitioners to make direct selections; Synthesis of strategies to contribute to the implementation of decentralized MLOps

Together, both syntheses allow us to expose homogeneity/heterogeneity across studies, as recommended by Kitchenham. We did not plan to conduct a quantitative synthesis as described by Kitchenham, since this would require comparable effect measures based on treatments and outcomes (which would require a lot of system context information that is not always provided equally across contributions). Given our RQs only target the identification and structuring of architectural approaches and supporting technologies rather than a comparison of the effectiveness of different approaches, we settled for the above synthesis described.

As with the data extraction, the final synthesis outcomes are going to be quality assured by the fourth researcher (Christian).